

lower node is  $0.6 \times 0.5 / 0.74 = 40.54\%$ . For the last two nodes at time 2, the final wealth for the buy-and-hold strategy is

$$\begin{aligned} W_{uu} &= 1.6400 \times (0.7317 \times 2.0 + 0.2683 \times 1.1) = 2.8840, \\ \text{or } W_{ud} &= 1.6400 \times (0.7317 \times 0.5 + 0.2683 \times 1.1) \\ &= 1.0840 = W_{du}, \end{aligned}$$

which is the same as  $W_{du} = 0.7400 \times (0.4054 \times 2.0 + 0.5946 \times 1.1)$

$$\begin{aligned} &= 1.0840 = W_{ud}, \\ \text{or } W_{dd} &= 0.7400 \times (0.4054 \times 0.5 + 0.5946 \times 1.1) = 0.6340. \end{aligned} \quad (4.19)$$

This is shown in Figure 4.9, Panel B at the end of the binomial tree.

Now consider the optimal rebalanced strategy, which rebalances at time 1 back to 60% equities and 40% bonds. The end of period wealth at time 1 is exactly the same as equation (4.18). The final wealth at time 2 for the rebalanced strategy is

$$\begin{aligned} W_{uu} &= 1.6400 \times (0.6 \times 2.0 + 0.4 \times 1.1) = 2.6896, \\ \text{or } W_{ud} &= 1.6400 \times (0.6 \times 0.5 + 0.4 \times 1.1) = 1.2136 = W_{du}, \\ \text{which is the same as } W_{du} &= 0.7400 \times (0.6 \times 2.0 + 0.4 \times 1.1) = 1.2136 = W_{ud} \\ \text{or } W_{dd} &= 0.7400 \times (0.6 \times 0.5 + 0.4 \times 1.1) = 0.5476. \end{aligned} \quad (4.20)$$

This set of payoffs is shown in Panel C of Figure 4.9.

The last panel D of Figure 4.9 plots the payoffs of the buy-and-hold strategy (equation (4.19)) and the rebalanced strategy (equation (4.20)) as a function of the stock value at maturity time 2. The buy-and-hold, un-rebalanced strategy is shown in the dashed straight line. The gains and losses on the buy-and-hold position are linear, by construction, in the stock price. The payoffs of the rebalanced strategy, in contrast, are *concave* over the stock price. Rebalancing adds more wealth to the investor if the stock price returns to 1.0 at maturity (1.2136 for the rebalanced vs. 1.0840 for the buy and hold for  $S_{ud} = S_{dd} = 1$ ). This is offset by the rebalancing strategy underperforming the buy-and-hold strategy when the ending stock values are low ( $S_{dd} = 0.25$ ) or high ( $S_{uu} = 4$ ).

This nonlinear pattern of the rebalancing strategy can be equivalently generated by short option positions. The strategy sells out-of-the-money call and put options and hence is short volatility.

Suppose there is a European call option with strike \$3.6760 maturing at time 2. This call option has the following payoffs at time 2:

$$\begin{aligned} C_{uu} &= \max(4.0000 - 3.6760, 0) = 0.3240, \\ \text{or } C_{ud} &= \max(1.0000 - 3.6760, 0) = 0 = C_{du}, \\ \text{or } C_{dd} &= \max(0.2500 - 3.6760, 0) = 0 \end{aligned} \quad (4.21)$$